Harshit Yadav

Machine Learning Engineer
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SUMMARY

Machine learning engineer with 3+ years building production agentic workflows, fintech decision systems, and ML/data platforms at Navi. Owns ambiguous operational problems from workflow discovery through controlled deployment, monitoring, evaluation, and stakeholder adoption.

EXPERIENCE

Navi Technologies, Bengaluru **July 2023 - Present**
Machine Learning Engineer II *Sept 2024 - Present*
Machine Learning Engineer I *July 2023 - Aug 2024*
Received the Navi Spearhead award within the 40-member Data Science organization.

On-Call Investigation & Remediation Automation

- Conceived and co-built a production LangGraph workflow that triages **100+ weekly alerts** and cut MTTR by approximately **90%** for automated alert classes versus historical manual baselines.
- Integrated Prometheus, Grafana, Kubernetes, Slack, GitHub, incident history, and service-owner SOPs; separated diagnostic/read tools from controlled remediation workflows, requiring human approval for novel failures and post-execution health validation before reuse.

Alternate-Data AI Underwriting Platform

- Co-designed and launched a production pilot for previously rejected personal-loan applicants using bureau, bank-statement, and SMS-derived signals; contributed to **INR 10-15 crore** in incremental monthly disbursals and a **36 bps** approval lift.
- Built a LangGraph decision workflow with meta-LLM aggregation, critic validation, and Qdrant RAG with citable evidence; incorporated Credit Risk feedback, isolated customer context, and reduced vendor costs by **INR 10-15 lakh per month**.

Agentic AI, ML Serving & Financial Data Platforms

- Built a production embedding-research agent that generates PyTorch pipelines, shortening research cycles by **85%** and increasing tested architectures by **12x**; added prompt versioning, structured outputs, LLM-as-judge checks, citation/data-quality validation, retries, fallbacks, access controls, and LangSmith monitoring.
- Built Cython/Triton inference for 100+ versioned models at **10 ms** and **100K RPM**, reducing AWS cost by **60%**; previously delivered a Go orchestrator with **80%** lower AWS cost and **95%** lower latency.
- Developed UPI rewards/referral pipelines and feature services over bureau, bank-statement, and SMS data, supporting credit-risk inference for personal-loan workflows at approximately **INR 100 crore in daily disbursals**.
- Established online and Databricks batch-scoring paths with self-service model onboarding; built a distributed Kotlin data-integration layer across gRPC, GraphQL, S3/Parquet, and Airflow.

Samsung Research Institute, Bengaluru **May 2022 - June 2022**
Software Developer Intern - Built a Hindi-language finance assistant using Rasa NLU and multilingual workflows; received a pre-placement offer.

Leoforce Technologies, Hyderabad **May 2019 - June 2019**
Full Stack Developer Intern - Built employee feedback and performance dashboards using JavaScript and SQL-backed services.

PROJECTS

Deep Researcher - Built a LangChain supervisor/worker research system with Tavily, private Qdrant RAG, function calling, JSON-schema outputs, and citations; parallel task delegation reduced model cost by **30x**.

SymPy Open-Source Contributions - Contributed the **DomainMatrix** module and enhancements across Matrix, Core, Polys, Derivatives, and Integrals in Python.

TECHNICAL SKILLS

AI/ML: Python, PyTorch, LangGraph, LangChain, LangSmith, RAG, Qdrant, prompt engineering, tool calling, structured outputs, LLM evaluation, human-in-the-loop controls

Data/Platforms: Databricks, Airflow, ETL/ELT, Kafka, S3, Parquet, Pandas, NumPy, SQL (working knowledge), PostgreSQL, Redis

Engineering: Go, Kotlin, Cython, C/C++, AWS, Kubernetes, Docker, Triton, Prometheus, Grafana, FastAPI, gRPC, GraphQL

EDUCATION

IIT Kharagpur - Integrated B.S./M.S., Mathematics and Computing **2018 - 2023 | 8/10**
CBSE Class 12 - SR Public Senior Secondary School 2018 | 94.3%
CBSE Class 10 - Atomic Energy Central School 2016 | 10/10

AWARDS

Navi Spearhead Award | INSPIRE Scholar | Top 1.7% in IIT-JEE Advanced 2018